

Product Requirements Document (PRD)

AI-Powered Academic Intelligence Platform (Working Name: Athena)

Version: 1.0

Status: Draft

Target Audience: Development Team, Product Team, UI/UX Designers

1. Project Overview

Vision

Build a modern AI-powered Learning Management System (LMS) designed specifically for colleges and universities. Unlike traditional LMS platforms that primarily store learning materials, this platform will function as an **Academic Intelligence Platform** that assists professors in creating courses, guides students through structured learning, and provides institutions with actionable academic insights.

The platform combines classroom management, AI-assisted course generation, guided learning, intelligent analytics, and academic performance monitoring into a single ecosystem.

Problem Statement

Current LMS platforms such as Moodle and Google Classroom primarily focus on content delivery. They do not actively help educators design better courses, identify struggling students, or improve academic outcomes.

Professors spend significant time creating learning materials, quizzes, and assessments manually, while administrators lack meaningful insights into classroom performance beyond grades and attendance.

This project aims to solve these problems using AI while ensuring that educators remain in complete control of academic content.

Product Goals

- Simplify course creation using AI.
 - Improve student engagement through guided learning.
 - Help professors identify weak concepts early.
 - Provide departments with actionable academic analytics.
 - Build an extensible platform suitable for universities and colleges.
-

2. User Roles

Super Admin

Responsible for platform-wide administration.

Responsibilities:

- Manage institutions
 - Manage system settings
 - User management
 - Platform analytics
 - Audit logs
 - Subscription management
-

Head of Department (HOD)

Responsible for monitoring departmental academic performance.

Features:

- Department dashboard
 - Class Health analytics
 - Faculty performance overview
 - Subject performance
 - Pass-rate predictions
 - AI-generated academic insights
 - Department reports
-

Class Teacher

Responsible for managing an entire classroom.

Permissions:

- Create classroom
- Manage students
- Assign subject teachers
- View complete classroom analytics
- View attendance
- View leaderboard
- Classroom announcements
- Overall student progress

The class teacher acts as the academic manager of the classroom.

Subject Teacher

Responsible only for their assigned subject.

Permissions:

- Upload syllabus
- Upload course materials
- Generate AI roadmap
- Edit AI-generated content
- Publish chapters
- Unlock chapters

- Create quizzes
- Grade assessments
- View subject-specific analytics

Subject teachers cannot modify other subjects.

Student

Features:

- Join classrooms
 - View assigned subjects
 - Study unlocked chapters
 - Complete quizzes
 - Submit assignments
 - Chat with AI Tutor
 - Review flashcards
 - Track personal progress
 - View leaderboard
-

3. Core Features

AI Course Builder

The professor uploads:

- Syllabus
- PDF notes
- PowerPoint presentations
- Lecture notes
- Previous question papers
- Additional learning resources

The AI automatically proposes:

- Course roadmap
- Learning milestones
- Units
- Chapters
- Learning objectives

- Flashcards
- Practice quizzes
- Assessments
- Estimated study time

The professor reviews, edits, and approves all generated content before publication.

Guided Learning

Learning is delivered in a structured manner rather than as a collection of files.

Features:

- Locked and unlocked chapters
- Progress tracking
- AI summaries
- Flashcards
- Practice quizzes
- Assignments
- Discussion board
- AI Tutor

Chapter unlocking is controlled by the professor to ensure the entire class progresses together.

Assessment System

Supports:

- MCQs
- Multiple Select
- True/False
- Fill in the Blanks
- Essay Questions

Features:

- Automatic grading
- Manual grading

- Question bank
 - Attempt history
 - Performance analytics
-

Leaderboards

Students earn points through:

- Quiz performance
- Assignment completion
- Participation
- Classroom engagement

Leaderboards encourage healthy competition without relying solely on speed.

4. AI Features

AI Tutor

Students can ask questions in natural language.

The AI retrieves answers from uploaded course material using Retrieval-Augmented Generation (RAG), ensuring responses remain aligned with the official curriculum.

AI Recommendations

For Students:

- Daily study suggestions
- Weak topic recommendations
- Revision reminders
- Personalized practice

For Professors:

- Students requiring intervention
- Weak classroom concepts
- Revision recommendations

- Engagement insights
-

Student Knowledge Graph

Rather than tracking only grades, the platform continuously measures concept mastery.

Example:

- Arrays – 95%
- Linked Lists – 90%
- Trees – 68%
- Graphs – 45%

This allows both students and professors to understand conceptual strengths and weaknesses.

AI Class Health Score

One of the platform's primary differentiators.

Instead of displaying only grades, AI calculates an overall classroom health score using:

- Attendance
- Quiz performance
- Assignment completion
- Student engagement
- Knowledge retention
- Learning progress

AI also generates actionable recommendations such as:

- Concepts requiring revision
 - Students at academic risk
 - Suggested intervention sessions
 - Predicted pass rate
-

AI Course Versioning

Every course maintains version history.

When a professor modifies:

- Roadmaps
- Chapters
- Quizzes
- Assessments
- Learning objectives

The platform records a new course version.

The AI also compares semesters and recommends improvements based on previous student performance.

Example:

- Split Unit 4 into two chapters
 - Add additional flashcards
 - Reduce quiz difficulty
 - Increase revision material
-

5. System Modules

Phase 1 (MVP)

- Authentication & RBAC
- Institution Management
- Classroom Management
- Subject Management
- Course Builder
- Learning Content
- Assignments
- Assessments
- Progress Tracking
- AI Course Builder
- AI Tutor
- AI Flashcards
- AI Quiz Generator
- AI Recommendations
- Analytics Dashboard

- AI Class Health Score
 - Student Knowledge Graph
 - Leaderboards
 - HOD Dashboard
 - AI Course Versioning
 - Reports
 - Notifications
-

Phase 2

Optional enhancements include:

- Academic Timeline
 - Mobile Application
 - Parent Portal
 - AI Voice Tutor
 - Placement Analytics
 - Calendar Integrations
 - Advanced AI Revision Sessions
-

6. Technology Stack

Frontend

- React 19
 - TypeScript
 - Vite
 - Tailwind CSS
 - shadcn/ui
 - TanStack Query
 - React Hook Form
 - Zod
-

Backend

- FastAPI
- Python 3.13+
- SQLAlchemy

- Alembic
 - Pydantic
 - Celery
 - Redis
-

Database

- PostgreSQL
 - pgvector
-

AI Platform

- OpenAI-compatible LLM Gateway
 - LangGraph
 - LlamaIndex
 - FastEmbed
 - Whisper
 - OCR Pipeline
 - RAG Engine
 - Prompt Management Service
-

Storage

- MinIO (Development)
 - Amazon S3 / Cloudflare R2 (Production)
-

Infrastructure

- Docker
 - Docker Compose
 - GitHub Actions
 - NGINX
 - Prometheus
 - Grafana
 - Sentry
-

7. Product Philosophy

This platform is designed around one guiding principle:

AI assists educators—it does not replace them.

Every AI-generated roadmap, assessment, quiz, flashcard, recommendation, and course revision must be reviewed and approved by a faculty member before becoming part of the official course.

The objective is not simply to digitize education but to improve learning outcomes through intelligent assistance, actionable analytics, and continuous academic improvement.

The long-term vision is to evolve from a traditional Learning Management System into a comprehensive **Academic Intelligence Platform** that helps educators teach more effectively, helps students learn more efficiently, and provides institutions with meaningful insights into academic success.